SPARK: Matrix-Gate Folding

A multilinear polynomial commitment IOP

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Abstract

We construct a family of linear codes for multilinear polynomials from GL_2 gate matrices attached to the nodes of a binary tree, a folding operator that is exactly the gate rule evaluated at a random point, and a commitment scheme in which the folding challenges are the opening point. The construction uses no structured evaluation domain: gates are arbitrary elements of $\mathrm{GL}_2(\mathbb{F})$ with nonzero second row, and the encoder uses only operations of the form $u + t\,v$. We prove the algebraic identities; three distance theorems for randomly sampled per-node gates (a one-layer bound, a rank-saturation bound, and a bound with a checked setup); that the generic code is MDS; a zero decomposition and an obstruction showing what any sharper distance bound must pay; a proximity lemma and an agreement lemma for the folding operator; and soundness of the commitment with a per-query detection rate of $\min(1-(1-\Delta)^{1/3}, \Delta/2)$, with no loss in the number of variables. The construction works over any field large enough for the distance bounds; gates may be sampled in an extension field. A Rust implementation accompanies this document.

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1 Notation

- $\mathbb{F}$ is a finite field with $q = |\mathbb{F}|$ (any characteristic, including 2); $\mathbb{K} \supseteq \mathbb{F}$ is an extension field used for verifier challenges.
- $\mathrm{ML}(x_j, \dots, x_n)$ denotes multilinear polynomials in the listed variables; ML_i denotes multilinear polynomials in i variables.
- $k \geq 0$ is the *redundancy parameter*. For $0 \leq i \leq n$ put

$$\Omega_i = \{0,1\}^k \times \{0,1\}^i, \qquad L_i = |\Omega_i| = 2^{k+i},$$

  and identify $\Omega_i = \Omega_{i-1} \times \{0,1\}$, writing $\omega = (\omega', b)$.
- A position $\omega \in \Omega_n$ is written $\omega = (\beta, \theta_n, \theta_{n-1}, \dots, \theta_1)$ with $\beta \in \{0,1\}^k$, and its prefixes are $\omega^{(j)} = (\beta, \theta_n, \dots, \theta_{j+1}) \in \Omega_{n-j}$ for $0 \leq j \leq n$.
- For $w, w' \in \mathbb{K}^{\Omega_i}$: $d_{\mathrm{pos}}(w, w') = L_i^{-1} \, |\{\omega : w(\omega) \neq w'(\omega)\}|$, and for $i \geq 1$ the *block distance*

$$d_{\mathrm{blk}}(w, w') = L_{i-1}^{-1} \left|\{\omega' \in \Omega_{i-1} : \ (w(\omega',0), w(\omega',1)) \neq (w'(\omega',0), w'(\omega',1))\}\right|.$$

  Always $d_{\mathrm{pos}} \leq d_{\mathrm{blk}} \leq 2 d_{\mathrm{pos}}$. Distances to a set are minima. For a linear code C, $\Delta(C)$ is its minimum relative distance.

2 Gates

Definition 2.1 (Gate)**.** A *gate* is a matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{GL}_2(\mathbb{F})$ with $c \neq 0$, $d \neq 0$. Write $\delta = ad - bc$ and $\mathcal{G}^* = \{A \in \mathrm{GL}_2(\mathbb{F}) : c \neq 0, \ d \neq 0\}$. Its *gate weights* are $W_L(x) = a + cx$ and $W_R(x) = b + dx$, and its *structural points* are

$$t^0 = -\frac{a}{c}, \qquad t^1 = -\frac{b}{d}.$$

Lemma 2.2 (Gates are point pairs with scales)**.** *The map* $(c, d, t^0, t^1) \mapsto \begin{pmatrix} -t^0 c & -t^1 d \\ c & d \end{pmatrix}$ *is a bijection from* $(\mathbb{F}^\times)^2 \times \{(t^0, t^1) \in \mathbb{F}^2 : t^0 \neq t^1\}$ *onto* $\mathcal{G}^*$*, and* $\delta = cd\,(t^1 - t^0)$*. In particular, if* A *is uniform on* $\mathcal{G}^*$ *then* (t^0, t^1) *is uniform on pairs of distinct elements.*

Proof. With $a = -t^0 c$, $b = -t^1 d$ one gets $ad - bc = -t^0 cd + t^1 cd = cd(t^1 - t^0)$, nonzero iff $t^0 \neq t^1$. The inverse map is $A \mapsto (c, d, -a/c, -b/d)$. The fibre over each (t^0, t^1) has size $(q-1)^2$. □

Lemma 2.3 (Decomposition and collapse)**.** *Let $A \in \mathcal{G}^*$ and let $F = u + x\,v$ with u, v independent of x. There are unique F_L, F_R independent of x with*

$$F = W_L(x)\,F_L + W_R(x)\,F_R, \qquad F_L = \delta^{-1}(du - bv), \quad F_R = \delta^{-1}(-cu + av).$$

Moreover

$$F\big|_{x=t^0} = \gamma^0 F_R, \quad \gamma^0 = -\frac{\delta}{c}, \qquad\qquad F\big|_{x=t^1} = \gamma^1 F_L, \quad \gamma^1 = +\frac{\delta}{d}.$$

Proof. $W_L F_L + W_R F_R = (aF_L + bF_R) + x(cF_L + dF_R)$, so the identity is $A\binom{F_L}{F_R} = \binom{u}{v}$, uniquely solvable since $\delta \neq 0$. At $x = t^0$: $W_L(t^0) = 0$ and $W_R(t^0) = b - ad/c = -\delta/c$. At $x = t^1$: $W_R(t^1) = 0$ and $W_L(t^1) = a - bc/d = \delta/d$. □

Lemma 2.4 (Gate rule at an arbitrary point)**.** *In the setting of Lemma 2.3, put $w^0 = F|_{x=t^0}$ and $w^1 = F|_{x=t^1}$. For every z,*

$$F\big|_{x=z} = W_L(z)\,\frac{d}{\delta}\,w^1 - W_R(z)\,\frac{c}{\delta}\,w^0 = w^0 + \frac{z - t^0}{t^1 - t^0}\,(w^1 - w^0).$$

Proof. By Lemma 2.3, $F_L = (d/\delta)w^1$ and $F_R = -(c/\delta)w^0$; substitute into $F = W_L F_L + W_R F_R$ at $x = z$. The second expression is the unique affine function of z taking the values w^0, w^1 at t^0, t^1. □

Lemma 2.5 (One-layer sum rule)**.** *If $F = W_L(x)F_L + W_R(x)F_R$ then $\sum_{x\in\{0,1\}} F = (2a + c)\,F_L + (2b + d)\,F_R$.*

Proof. $\sum_{x\in\{0,1\}}(a + cx) = 2a + c$ and $\sum_{x\in\{0,1\}}(b + dx) = 2b + d$. □

3 The gate code

Definition 3.1 (Gate family)**.** A *gate family* is a collection $\mathcal{A} = \big(A_j(\omega')\big)_{1\le j\le n,\ \omega'\in\Omega_{n-j}}$ of gates, one per variable x_j and per node $\omega' \in \Omega_{n-j}$. Write $t_j^b(\omega')$ for the structural points of $A_j(\omega')$. The family is *per-layer* if $A_j(\omega')$ does not depend on ω', and *per-node* otherwise.

Definition 3.2 (Encoding)**.** For $0 \le i \le n$ let C_i encode $\mathrm{ML}(x_{n-i+1}, \dots, x_n)$ as follows.

$$\mathrm{Enc}_0(c)(\beta) = c \quad (\beta \in \{0,1\}^k),$$

and for $i \ge 1$, writing $j = n - i + 1$ and $F = u + x_j\,v$ with $u, v \in \mathrm{ML}(x_{j+1}, \dots, x_n)$,

$$\mathrm{Enc}_i(F)(\omega', b) = \mathrm{Enc}_{i-1}(u)(\omega') + t_j^b(\omega')\,\mathrm{Enc}_{i-1}(v)(\omega'), \qquad \omega' \in \Omega_{i-1},\ b \in \{0,1\}.$$

Put $C_i = \mathrm{Enc}_i(\mathrm{ML}(x_{n-i+1}, \dots, x_n)) \subseteq \mathbb{F}^{\Omega_i}$.

Proposition 3.3 (Closed form)**.** *For $F \in \mathrm{ML}(x_1, \dots, x_n)$ and $\omega = (\beta, \theta_n, \dots, \theta_1) \in \Omega_n$,*

$$\mathrm{Enc}_n(F)(\omega) = F\big(r(\omega)\big), \qquad r_j(\omega) = t_j^{\theta_j}\big(\omega^{(j)}\big).$$

The coordinate $r_j(\omega)$ depends only on $(\beta, \theta_n, \dots, \theta_j)$. Hence two positions that differ only in θ_j have points differing only in the coordinate x_j.

Proof. Induction on n. For $F = u + x_1 v$, $\mathrm{Enc}_n(F)(\omega^{(1)}, \theta_1) = u(r_{\ge 2}) + t_1^{\theta_1}(\omega^{(1)})\,v(r_{\ge 2}) = F(r(\omega))$, where $r_{\ge 2}$ is the point attached to $\omega^{(1)}$ by the induction hypothesis. □

Remark 3.4 (Relation to the bottom table)**.** By Lemma 2.3, the bottom table T_0 of the original symbolic folding protocol satisfies $T_0[\theta] = \Gamma(\theta)^{-1}F(r(\theta))$ with a public nonzero $\Gamma(\theta)$. So T_0 is $\mathrm{Enc}_n(F)$ for $k = 0$ and a per-layer family, up to a public diagonal scaling, which preserves all distances. Propositions 3.7 and 3.8 explain why that table alone cannot support spot-checks.

Proposition 3.5 (Encoding cost)**.** $\mathrm{Enc}_n(F)$ *is computed with at most* $(n+1)\,2^{n+k}$ *multiplications and as many additions in* $\mathbb{F}$*.*

Proof. With $T(i)$ the cost for i variables, $T(i) = 2T(i-1) + L_i$ and $T(0) = L_0 = 2^k$, so $T(n) = (n+1)2^{n+k}$. □

Lemma 3.6 (Extension of scalars)**.** *For a linear code* $C \subseteq \mathbb{F}^N$*, the code* $C \otimes \mathbb{K} \subseteq \mathbb{K}^N$ *has* $\Delta(C \otimes \mathbb{K}) = \Delta(C)$*.*

Proof. Fix an $\mathbb{F}$-basis (e_l) of $\mathbb{K}$. A word of $C \otimes \mathbb{K}$ is $\sum_l e_l c_l$ with $c_l \in C$. At a position p, $\sum_l e_l c_l(p) = 0$ iff all $c_l(p) = 0$, so the support is $\bigcup_l \operatorname{supp} c_l$. □

Proposition 3.7 (Per-layer families have distance 2^{-n})**.** *If* $\mathcal{A}$ *is per-layer, then* $\Delta(C_n) \le 2^{-n}$*.*

Proof. All points $r(\omega)$ lie in the grid $\prod_j \{t_j^0, t_j^1\}$, each grid point appearing 2^k times. $F = \prod_{j=1}^n (x_j - t_j^0)$ is nonzero exactly at the grid point $(t_1^1, \dots, t_n^1)$, so $\mathrm{wt}(\mathrm{Enc}_n(F)) = 2^k$, of relative weight 2^{-n}. □

Proposition 3.8 (No redundancy, no testing)**.** *If* $k = 0$ *and* $\Delta(C_n) > 0$*, then* $C_n = \mathbb{F}^{\Omega_n}$*. Every table is then a codeword, and* $\Delta(C_n) = 2^{-n}$*.*

Proof. Enc_n is injective and $\dim C_n = 2^n = L_n$. □

3.1 Distance of per-node random families

Lemma 3.9 (One-layer distance)**.** *Let* $C \subseteq \mathbb{F}^N$ *be linear with* $\dim C = D$ *and* $\Delta(C) \ge \Delta$*. Let* $(t_p^0, t_p^1)_{p \le N}$ *be independent, each uniform on pairs of distinct elements of* $\mathbb{F}$*, and*

$$C' = \big\{(c_u + t^0 \circ c_v,\ c_u + t^1 \circ c_v)\ :\ c_u, c_v \in C\big\} \subseteq \mathbb{F}^{2N},$$

with coordinates paired as $(p,0),(p,1)$*. For every integer* $s \ge 0$*,*

$$\Pr\Big[\Delta(C') < \Delta - \tfrac{s}{2N} \quad \text{or} \quad \dim C' < 2D\Big] \ \le\ q^{2D}\binom{N}{s}\Big(\frac{2}{q}\Big)^s.$$

Proof. Fix $(c_u, c_v) \neq (0,0)$ and let w be the corresponding word of C'.

If $c_v = 0$, then $w = (c_u, c_u)$ and $\mathrm{wt}(w) = 2\,\mathrm{wt}(c_u) \ge 2\Delta N$.

If $c_v \neq 0$, consider $p \in \operatorname{supp} c_v$. The pair $(c_u(p) + t_p^0 c_v(p),\ c_u(p) + t_p^1 c_v(p))$ has at most one zero, since $t_p^0 \neq t_p^1$. It has a zero iff $\alpha_p := -c_u(p)/c_v(p) \in \{t_p^0, t_p^1\}$, an event of probability exactly $2/q$, independently over p. Let Z be the set of such p. Then $\mathrm{wt}(w) \ge 2\,\mathrm{wt}(c_v) - |Z| \ge 2\Delta N - |Z|$, and $\Pr[|Z| \ge s] \le \binom{N}{s}(2/q)^s$.

A union bound over the q^{2D} pairs gives: outside the stated event every nonzero pair has relative weight $\ge \Delta - s/(2N) > 0$ (when this is positive), which also gives injectivity, hence $\dim C' = 2D$. □

Theorem 3.10 (Distance of the gate code)**.** *Let $\mathcal{A}$ be a per-node family with independent gates, each uniform on $\mathcal{G}^*$. For integers $s_1, \dots, s_n \geq 0$, with probability at least*

$$1 - \sum_{i=1}^{n} q^{2^i} \binom{L_{i-1}}{s_i} \left(\frac{2}{q}\right)^{s_i},$$

every C_i $(0 \leq i \leq n)$ has dimension 2^i and

$$\Delta(C_i) \;\geq\; \Delta^* := 1 - \sum_{l=1}^{n} \frac{s_l}{2L_{l-1}}.$$

Proof. C_0 is the repetition code, with $\Delta(C_0) = 1$ and dimension 1. Definition 3.2 builds C_i from C_{i-1} exactly as C' from C in Lemma 3.9, with $N = L_{i-1}$, $D = 2^{i-1}$ and fresh structural points (uniform distinct pairs by Lemma 2.2). Apply Lemma 3.9 at each level and take a union bound. □

Remark 3.11 (Parameters)**.** Using $\binom{N}{s} \leq (eN/s)^s$, Table 1 lists the smallest certified Δ^* with setup failure probability $\leq 2^{-128}$. Sample gates in the base field $\mathbb{F}$. By Lemma 3.6, the distance carries over to $\mathbb{K}$.

$\log_2 q$	n	k	certified Δ^*
31	16	5	0.212
31	16	6	0.589
31	20	6	0.508
31	24	6	0.427
62	20	6	0.604
124	20	5	0.287
124	20	6	0.640

Table 1: Certified distance (Theorem 3.10), setup failure $\leq 2^{-128}$. For $k \leq 4$ the certificate is vacuous at these n.

q	n	k	length	per-node min	per-node mean	per-layer
13	2	3	32	0.563	0.625	0.250
61	2	3	32	0.781	0.797	–
7	3	2	32	0.250	0.339	0.125
7	3	3	64	0.375	0.412	0.125
3	4	2	64	0.078	0.096	0.0625

Table 2: Exhaustive minimum relative distance over random families (6 samples, 4 for $q = 61$). Per-layer families sit exactly at 2^{-n} (Proposition 3.7). Per-node distance grows with q, as Lemma 3.9 predicts ($2/q$ collision rate).

3.2 Distance by rank saturation

Theorem 3.10 loses about 2^{-k} per level. The argument below loses only about $H_2(2^{-k})/\log_2 q$ per level, so large fields allow small redundancy.

For a set X of positions of C_i write $V_X = \{F : \mathrm{Enc}_i(F)|_X = 0\}$ and $d_i(X) = \dim V_X$.

Lemma 3.12 (Augmentation)**.** *Let $C = \mathrm{Enc}(\mathrm{ML})$ be injective and let $Z \geq 1$. If $V_S = 0$ for every S with $|S| \geq Z$, then $d(X) \leq \max(0, Z - |X|)$ for every X.*

Proof. Suppose $d(X) = d \geq 1$. Since Enc is injective, some element of V_X is nonzero at some position ω, and $d(X \cup \{\omega\}) = d-1$. Repeating $d-1$ times gives Y with $|Y| = d-1$ and $V_{X \cup Y} \neq 0$. Hence $|X| + d - 1 < Z$. □

Lemma 3.13 (One level)**.** *Fix the gates of C_{i-1}, and assume $d_{i-1}(X) \leq \max(0, Z' - |X|)$ for all X. Let $S \subseteq \Omega_i$ with $|S| \geq 2Z'$. Over the level-i gates,*

$$\Pr\big[V_S \neq 0\big] \;\leq\; \frac{(1+1/q)^{L_{i-1}}}{q-1}\, q^{2Z'-|S|}.$$

Proof. Let T_2 be the blocks with both entries in S and T_1 the blocks with exactly one, say $(\omega', b_{\omega'})$; then $|S| = 2|T_2| + |T_1|$. Write $F = u + xv$ and $U = V_{T_2}$ (level $i-1$), and put $\hat{u} = \mathrm{Enc}_{i-1}(u)$, $\hat{v} = \mathrm{Enc}_{i-1}(v)$.

Constraints. $F \in V_S$ iff:

- $u, v \in U$: on a double block, $\hat{u} + t^0\hat{v} = \hat{u} + t^1\hat{v} = 0$ with $t^0 \neq t^1$ forces $\hat{u} = \hat{v} = 0$;
- for every $\omega' \in T_1$: $\hat{u}(\omega') + t\,\hat{v}(\omega') = 0$, where $t = t^{b_{\omega'}}(\omega')$ is uniform on $\mathbb{F}$ and independent across blocks.

For fixed (u, v), the constraint at $\omega' \in T_1$ holds with probability $1/q$ if $\hat{v}(\omega') \neq 0$, and with probability $[\hat{u}(\omega') = 0]$ if $\hat{v}(\omega') = 0$.

Counting. V_S is a subspace, so $\Pr[V_S \neq 0] \leq \mathbb{E}[\#(V_S \setminus 0)]/(q-1)$.

- Pairs with $v = 0$ contribute $q^{d_{i-1}(T_2 \cup T_1)} - 1$. This is 0, because $|T_2| + |T_1| \geq |S|/2 \geq Z'$.
- For $v \neq 0$, let $N(v) = \{\omega' \in T_1 : \hat{v}(\omega') = 0\}$. Summing over u gives the weight $q^{|N(v)|-|T_1|}\, q^{d_{i-1}(T_2 \cup N(v))}$.
- Group the $v \neq 0$ by $N = N(v)$. They lie in $V_{T_2 \cup N} \setminus 0$, which is nonempty only if $d_{i-1}(T_2 \cup N) \geq 1$, i.e. $|T_2| + |N| < Z'$. Then $d_{i-1}(T_2 \cup N) \leq Z' - |T_2| - |N|$.

Hence

$$\mathbb{E}[\#(V_S \setminus 0)] \leq \sum_{N \subseteq T_1} q^{|N|-|T_1|+2(Z'-|T_2|-|N|)} = q^{2Z'-|S|} \sum_{N \subseteq T_1} q^{-|N|} = q^{2Z'-|S|}(1+1/q)^{|T_1|}. \quad \square$$

Theorem 3.14 (Distance by rank saturation)**.** *Let the gates be independent and uniform on $\mathcal{G}^*(\mathbb{F})$. Choose integers $g_1, \ldots, g_n \geq 0$, put $Z_0 = 1$ and $Z_i = 2Z_{i-1} + g_i$, and assume $Z_i \leq L_i$. With probability at least*

$$1 - \sum_{i=1}^{n} \binom{L_i}{Z_i} \frac{(1+1/q)^{L_{i-1}}}{q-1}\, q^{-g_i},$$

every nonzero codeword of every C_i has fewer than Z_i zeros. In particular $\Delta(C_i) \geq 1 - (Z_i - 1)/L_i$, and $\min_i \Delta(C_i) \geq 1 - (Z_n - 1)/L_n$.

Proof. Induction on i.

- *Base.* C_0 is the repetition code, so $V_S = 0$ for every nonempty S, i.e. $Z_0 = 1$.
- *Step.* Assume the statement for C_{i-1}. Then Enc_{i-1} is injective (because $Z_{i-1} \leq L_{i-1}$), and Lemma 3.12 gives $d_{i-1}(X) \leq \max(0, Z_{i-1} - |X|)$ for all X.

- *Union bound.* By Lemma 3.13 with $Z' = Z_{i-1}$, each S with $|S| = Z_i$ satisfies $V_S \neq 0$ with probability at most $\frac{(1+1/q)^{L_{i-1}}}{q-1} q^{-g_i}$. A union bound over the $\binom{L_i}{Z_i}$ such sets, together with monotonicity in S, gives $V_S = 0$ for all $|S| \geq Z_i$.

The excess $(Z_i - D_i)/L_i$ is nondecreasing in i and $1/L_i$ decreases, so C_n attains the minimum. □

Remark 3.15 (Size of the loss)**.** Writing $Z_i = D_i + e_i$ gives $e_i = 2e_{i-1} + g_i$, so $e_i/L_i = e_{i-1}/L_{i-1} + g_i/L_i$. The union bound needs $g_i \log_2 q \gtrsim L_i H_2(Z_i/L_i) + \lambda$. So the relative loss per level is about $H_2(2^{-k})/\log_2 q$, plus $\lambda/(L_i \log_2 q)$ on the first few levels. Gates may be sampled in an extension $\mathbb{F}' \supseteq \mathbb{F}$, at the cost of storing E_0 in $\mathbb{F}'$.

gate field	n	copies	Δ (Thm. 3.14)	queries	Δ (Thm. 3.10)
Goldilocks (2^{64})	20	16	0.579	327	< 0
Goldilocks (2^{64})	20	32	0.729	218	0.221
Goldilocks (2^{64})	20	64	0.811	183	0.605
Goldilocks^3 (2^{192})	20	4	0.675	253	< 0
Goldilocks^3 (2^{192})	20	8	0.812	183	< 0
M31 (2^{31})	20	64	0.417	523	0.508

Table 3: Certified distance with setup failure $\leq 2^{-128}$, and the resulting queries for 2^{-128} soundness (Theorem 6.4, $\delta = 0.95\,\delta_{\max}$). Both theorems are valid, so the larger value applies.

Theorem 3.16 (The generic code is MDS)**.** *Regard the structural points $t_i^b(\omega')$ as independent indeterminates, and work over the field of rational functions in them. Then every set of at most D_i coordinate functionals of C_i is linearly independent; equivalently, the coordinate matroid of C_i is uniform of rank D_i, and the generic code C_n is MDS.*

Proof. Induction on i. For $i = 0$ every coordinate functional is $c \mapsto c \neq 0$.

Step. Identify ML_i with ML_{i-1}^2 via $F = u + xv$. Position (ω', b) has functional $(\ell_{\omega'}, t^b \ell_{\omega'})$, where $\ell_{\omega'}$ is the coordinate functional of C_{i-1} at ω'. Let S be a set of positions with $|S| \leq 2D'$, $D' = D_{i-1}$. Let A be its double blocks and B its single blocks, with $a = |A|$ and $b = |B|$, so $2a + b \leq 2D'$.

Double blocks. Since $t^1 - t^0 \neq 0$, the two rows of a double block span $(\ell_{\omega'}, 0)$ and $(0, \ell_{\omega'})$. By induction the $\ell_{\omega'}$, $\omega' \in A$, are independent, because $a \leq D'$. So the double blocks contribute rank $2a$ and span $U \oplus U$, with $U = \mathrm{span}\{\ell_{\omega'} : \omega' \in A\}$.

Quotient. Pass to $W = \mathrm{ML}_{i-1}^*/U$, of dimension $m = D' - a$, and write $\bar{\ell}_{\omega'}$ for the images. By induction, any at most m of the $\bar{\ell}_{\omega'}$, $\omega' \in B$, are independent: if $|T| \leq m$ then $|A \cup T| \leq D'$.

Single blocks. Since $b \leq 2m$, split $B = B_0 \sqcup B_1$ with $|B_0|, |B_1| \leq m$. Choose coordinates J_0 and J_1 of W with $|J_r| = |B_r|$ such that the minors $\det M_{B_0}$ and $\det M_{B_1}$ of $\{\bar{\ell}_{\omega'}\}$ are nonzero.

Consider the $b \times b$ minor of the vectors $(\bar{\ell}_{\omega'}, t_{\omega'} \bar{\ell}_{\omega'})$ on the columns J_0 (first copy of W) and J_1 (second copy). By the Laplace expansion along the column split (J_0, J_1), the monomial $\prod_{\omega' \in B_1} t_{\omega'}$ comes only from the term that puts the rows B_1 on J_1. Its coefficient is $\pm \det M_{B_0} \det M_{B_1} \neq 0$, and it does not involve the level-i variables. So the minor is nonzero and the b vectors are independent modulo $U \oplus U$.

The total rank is $2a + b = |S|$. □

Corollary 3.17 (Very large fields)**.** *If $q - 1 \geq 2^\lambda n D \binom{L}{D}$ (with $D = D_n$, $L = L_n$), then with probability at least $1 - 2^{-\lambda}$ over uniform gates, C_n is MDS: $\Delta(C_n) = 1 - (D-1)/L$.*

Proof. Parametrise each pair as $(t^0, t^1) = (a, a+s)$ with $(a, s) \in \mathbb{F} \times \mathbb{F}^\times$. This is a bijection onto distinct pairs, and an invertible linear change of variables.

For $|S| = D$, Theorem 3.16 makes $\det M_S$ a nonzero polynomial. Each entry of M_S is a product of at most n structural points, so $\deg \det M_S \le nD$. Schwartz–Zippel over $\mathbb{F} \times \mathbb{F}^\times$ gives $\Pr[\det M_S = 0] \le nD/(q-1)$. Take a union bound over the $\binom{L}{D}$ sets. □

Remark 3.18. Corollary 3.17 needs $\log_2 q \gtrsim L\, H_2(R)$, so it is structural rather than practical. It shows that the level count in Theorem 3.14 is not forced by the construction. Over Goldilocks with random gates, every D-subset was checked to be nonsingular for $(n, k) \in \{(2,1), (2,2), (2,3), (3,1), (3,2)\}$ (about $1.06 \cdot 10^7$ subsets).

A per-set bound cannot close the gap: for $|S| = D+1$ the measured $\Pr[\operatorname{rank} M_S < D]$ is about $1.5/q$, not about q^{-2}, because a single coincidence of gate points makes many sets degenerate at once.

3.3 Checked setup

The losses of Theorem 3.14 at the first levels come from short codes, where a union bound is expensive. These levels are small enough to be verified once, at setup.

Theorem 3.19 (Distance with a checked setup)**.** *Fix $i_0 \ge 0$. Sample the gates of levels $\le i_0$ (variables $x_n, \dots, x_{n-i_0+1}$) and resample them until C_{i_0} is MDS. Sample the gates of levels $> i_0$ independently and uniformly on $\mathcal{G}^*(\mathbb{F})$. Choose $g_{i_0+1}, \dots, g_n \ge 0$, put $Z_{i_0} = D_{i_0}$ and $Z_i = 2Z_{i-1} + g_i$, and assume $Z_i \le L_i$. With probability at least*

$$1 - \sum_{i=i_0+1}^{n} \binom{L_i}{Z_i} \frac{(1+1/q)^{L_{i-1}}}{q-1}\, q^{-g_i},$$

every nonzero codeword of C_n has fewer than Z_n zeros, so $\Delta(C_n) \ge 1 - (Z_n - 1)/L_n$.

Proof. C_{i_0} is MDS, so every nonzero codeword of C_{i_0} has at most $D_{i_0} - 1$ zeros: $V_S = 0$ whenever $|S| \ge Z_{i_0}$. The acceptance test depends only on the gates of levels $\le i_0$, so the gates of levels $> i_0$ are still independent and uniform after conditioning on acceptance. Run the induction of Theorem 3.14 from $i = i_0 + 1$. □

Remark 3.20 (Cost of the check)**.** C_1 is MDS iff the 2^{k+1} points of the level-1 gates are pairwise distinct. C_2 is MDS iff every 4-subset of its 2^{k+2} positions gives a nonsingular 4×4 matrix with rows $(1, y_1, y_2, y_1 y_2)$. By Theorem 3.16 and Schwartz–Zippel, a uniform family is rejected with probability at most $\binom{L_{i_0}}{D_{i_0}}\, i_0 D_{i_0}/(q-1)$, which is negligible for $i_0 \le 2$ at $q \approx 2^{64}$. The check is done once; the verifier only recomputes the gates from the public seed and the accepted counter.

gate field	copies	$i_0 = 0$	$i_0 = 1$	$i_0 = 2$	$i_0 = 3$
M31 (2^{31})	64	0.417	0.527	0.617	0.683
Goldilocks (2^{64})	16	0.579	0.669	0.742	0.768
Goldilocks (2^{64})	32	0.729	0.789	0.822	0.854
Goldilocks (2^{64})	64	0.811	0.865	0.888	0.905

Table 4: Certified $\Delta(C_n)$ from Theorem 3.19, $n = 20$, setup failure $\le 2^{-128}$. The column $i_0 = 0$ is Theorem 3.14.

$\log_2 q$ (gate field)	copies	Δ	$1-R$	queries
64	16	0.742	0.938	210
64	32	0.822	0.969	180
128	8	0.757	0.875	201
128	16	0.863	0.938	169
256	8	0.836	0.875	176
256	16	0.913	0.938	157

Table 5: Gates in a larger field (e.g. an extension of the coefficient field), $i_0 = 2$, $n = 20$, $\lambda = 128$. Queries for 2^{-128} soundness by Theorem 6.4 with $\delta = 0.95\,\delta_{\max}$. A larger gate field reduces the number of copies; the number of queries is bounded below by the radius $\delta < \Delta/2$ of Lemma 6.2.

3.4 Zero structure

For $F \in \mathrm{ML}(x_1, \dots, x_n)$ and a prefix $c = (\beta, \theta_n, \dots, \theta_{n-i+1}) \in \Omega_i$, let $F_c \in \mathrm{ML}(x_1, \dots, x_{n-i})$ be F with x_l replaced by $r_l(c)$ for $l > n - i$ (Proposition 3.3). The prefix c is *killed* if $F_c = 0$, and *maximally killed* if moreover its parent is not killed.

Lemma 3.21 (Zero decomposition)**.** *Let $F \neq 0$ and let $\kappa_i(F)$ be the number of maximally killed prefixes in Ω_i. The zero set of $\mathrm{Enc}_n(F)$ is the disjoint union of the subtrees below the maximally killed prefixes, and*

$$\#\{\omega : \mathrm{Enc}_n(F)(\omega) = 0\} = \sum_{i=1}^{n} \kappa_i(F)\, 2^{n-i}.$$

Moreover, if $p \in \Omega_{i-1}$ is not killed, at most one child of p is killed. For fixed F and fixed gates of the levels above p, a given child of p is killed with probability at most $1/q$ over the gate at p.

Proof. By Proposition 3.3, $\mathrm{Enc}_n(F)(\omega) = F_\omega$, so the zeros are the killed leaves. On the path from the root to a zero leaf, $F_{(\beta)} = F \neq 0$ and the leaf is killed, so there is a first killed prefix; it is maximally killed. Every prefix below a killed prefix is killed, since a restriction of 0 is 0. Two maximally killed prefixes are not comparable, so their subtrees are disjoint, and a prefix in Ω_i has 2^{n-i} leaves.

Write $F_p = A + xB$ with $x = x_{n-i+1}$. The children are $A + t^0(p)B$ and $A + t^1(p)B$. If both vanish, then $(t^0(p) - t^1(p))B = 0$, so $B = 0$, $A = 0$ and $F_p = 0$. A child is killed iff $A = -t\,B$ for its point t: this is impossible if $B = 0$ (then $A \neq 0$), and otherwise holds for at most one value of t. □

Proposition 3.22 (Coincidences cost $R\,2^{-j}$)**.** *Let $1 \le j \le n$ and let $p \neq p' \in \Omega_{j-1}$ lie in different copies ($\beta \neq \beta'$). If two structural points of their gates coincide, $t^b(p) = t^{b'}(p') = s$, then*

$$\Delta(C_n) \;\le\; 1 - R\,(1 + 2^{-j}) + \frac{1}{L_n}.$$

Proof. Write $p = (\beta, \theta_n, \dots, \theta_{n-j+2})$. For $1 \le i \le j-1$ let c_i be the structural point of the sibling, off the path of p, at level i: $c_i = t_{n-i+1}^{1-\theta_{n-i+1}}(\beta, \theta_n, \dots, \theta_{n-i+2})$. Put

$$F = \prod_{i=1}^{j-1} \big(x_{n-i+1} - c_i\big) \cdot \big(x_{n-j+1} - s\big) \cdot K(x_1, \dots, x_{n-j}),$$

which is multilinear. It vanishes on:

- the $j-1$ sibling subtrees off the path of p, of sizes $2^{n-1},\dots,2^{n-j+1}$, which sum to 2^n-2^{n-j+1};
- the subtrees below (p,b) and (p',b'), of size 2^{n-j} each. They are disjoint from the previous ones ((p,b) is on the path of p, and p' is in another copy).

This gives 2^n zeros. The factor K ranges over a space of dimension 2^{n-j}, so it can be chosen nonzero and vanishing at $2^{n-j}-1$ further positions. Then $F\neq 0$ has at least $2^n+2^{n-j}-1 = D_n-1+2^{n-j}$ zeros, a relative weight of at most $1-R-R\,2^{-j}+1/L_n$. □

Remark 3.23 (Consequences for the level count)**.** Each pair of points of distinct level-j nodes coincides with probability $1/q$. So a theorem with failure probability $2^{-\lambda} < 1/q$ must allow a loss of order $(\lambda/\log_2 q)\,R\,2^{-i_0}$ after a checked setup at level i_0. Coincidences are not the only source of excess zeros: for $n=2$, four positions in different level-2 nodes whose points lie on a line $x_2 = a+bx_1$ give the nonzero $F = x_2-a-bx_1$ with $4=D_2$ zeros, although all points are distinct. A proof that removes the level count must therefore control algebraic relations among many gate points at once.

Open problem 3.24 (Removing the level count)**.** Theorems 3.14 and 3.19 lose about $H_2(2^{-k})/\log_2 q$ per level. Prove a bound

$$\Delta(C_n) \;\geq\; 1-R-c_1\frac{H_2(R)}{\log_2 q}-c_2\frac{\lambda}{\log_2 q}\,R\,2^{-i_0}-o(1)$$

with c_1,c_2 independent of n, or show that a loss growing with n is unavoidable.

4 Folding

Definition 4.1 (Fold)**.** For $1\leq j\leq n$ and $z\in\mathbb{K}$, define $\mathrm{fold}_{j,z}:\mathbb{K}^{\Omega_{n-j+1}}\to\mathbb{K}^{\Omega_{n-j}}$ by

$$(\mathrm{fold}_{j,z}w)(\omega') = w(\omega',0)+\lambda_{j,\omega'}(z)\big(w(\omega',1)-w(\omega',0)\big),\qquad \lambda_{j,\omega'}(z)=\frac{z-t_j^0(\omega')}{t_j^1(\omega')-t_j^0(\omega')}.$$

By Lemma 2.4, this equals the gate rule of $A_j(\omega')$ at $x_j=z$:

$$(\mathrm{fold}_{j,z}w)(\omega') = W_L(z)\,\tfrac{d}{\delta}\,w(\omega',1)-W_R(z)\,\tfrac{c}{\delta}\,w(\omega',0).$$

Lemma 4.2 (Compatibility)**.** *For $F\in\mathrm{ML}(x_j,\dots,x_n)$,*

$$\mathrm{fold}_{j,z}\big(\mathrm{Enc}_{n-j+1}(F)\big)=\mathrm{Enc}_{n-j}\big(F|_{x_j=z}\big).$$

Consequently, folding $\mathrm{Enc}_n(F)$ with $z_1,\dots,z_n$ gives the constant word $F(z_1,\dots,z_n)$.

Proof. With $F=u+x_jv$, the block at ω' is $(e_u+t^0e_v,\ e_u+t^1e_v)$, where $e_u=\mathrm{Enc}(u)(\omega')$ and $e_v=\mathrm{Enc}(v)(\omega')$. Its fold is $e_u+z\,e_v$, and $\mathrm{Enc}(u)+z\mathrm{Enc}(v)=\mathrm{Enc}(u+zv)$ by linearity. □

Lemma 4.3 (Contraction)**.** $d_{\mathrm{pos}}(\mathrm{fold}_{j,z}w,\mathrm{fold}_{j,z}w')\leq d_{\mathrm{blk}}(w,w')$.

Proof. $(\mathrm{fold}w)(\omega')$ depends only on the block of w at ω'. □

Lemma 4.4 (Line form)**.** *For $w \in \mathbb{K}^{\Omega_{n-j+1}}$ define $A, B \in \mathbb{K}^{\Omega_{n-j}}$ by gate inversion:*

$$B(\omega') = \frac{w(\omega', 1) - w(\omega', 0)}{t_j^1(\omega') - t_j^0(\omega')}, \qquad A(\omega') = w(\omega', 0) - t_j^0(\omega')B(\omega').$$

Then $\mathrm{fold}_{j,z} w = A + zB$ for all z. The map $w \mapsto (A, B)$ is bijective. For $a, b \in C_{n-j}$, the word w agrees with the codeword of C_{n-j+1} built from (a, b) on exactly the blocks ω' where $(A, B)(\omega') = (a, b)(\omega')$.

Proof. Direct from Definition 3.2 and $t^0 \neq t^1$. □

Lemma 4.5 (Line proximity)**.** *Let $C \subseteq \mathbb{K}^N$ be linear with $\Delta(C) \geq \Delta$, let $0 \leq \delta < \Delta/3$, and $A, B \in \mathbb{K}^N$. Let $\mathcal{Z} = \{z \in \mathbb{K} : d_{\mathrm{pos}}(A + zB, C) \leq \delta\}$. If $|\mathcal{Z}| = M \geq 2$, there exist $a, b \in C$ with*

$$\big|\{p : (A(p), B(p)) \neq (a(p), b(p))\}\big| \;\leq\; \frac{M}{M-1}\,\delta N.$$

Proof. Take $z_1 \neq z_2$ in $\mathcal{Z}$, with codewords c_1, c_2 agreeing with $A + z_i B$ on sets S_1, S_2 of size $\geq (1-\delta)N$. Put $b = (c_1 - c_2)/(z_1 - z_2)$ and $a = c_1 - z_1 b$, both in C. On $S_1 \cap S_2$ we have $A = a$ and $B = b$.

For any $z_3 \in \mathcal{Z}$ with codeword c_3 and agreement set S_3, the codewords c_3 and $a + z_3 b$ agree on $S_1 \cap S_2 \cap S_3$. This set has size $\geq (1 - 3\delta)N > (1 - \Delta)N$, so $c_3 = a + z_3 b$. Hence every $z \in \mathcal{Z}$ has agreement set $S_z \subseteq \{p : A(p) + zB(p) = a(p) + zb(p)\}$.

Let $E = \{p : (A, B)(p) \neq (a, b)(p)\}$. For $p \in E$ the affine function $z \mapsto (A - a)(p) + z(B - b)(p)$ is nonzero, so $p \in S_z$ for at most one z. Then $\sum_{z \in \mathcal{Z}} |E \cap S_z| \leq |E|$. Combined with $|E \setminus S_z| \leq \delta N$, this gives $M|E| - |E| \leq M\delta N$. □

Corollary 4.6 (Folding preserves distance)**.** *Let $\Delta_{n-j} = \Delta(C_{n-j})$, $0 \leq \delta < \Delta_{n-j}/3$ and $\eta > 0$. If $d_{\mathrm{blk}}(w, C_{n-j+1} \otimes \mathbb{K}) \geq (1 + \eta)\delta$, then*

$$\big|\{z \in \mathbb{K} :\; d_{\mathrm{pos}}(\mathrm{fold}_{j,z} w,\; C_{n-j} \otimes \mathbb{K}) \leq \delta\}\big| \;\leq\; 1 + \eta^{-1}.$$

Proof. Write $\mathrm{fold}_{j,z} w = A + zB$ (Lemma 4.4) and let M be the number of such z. If $M \geq 2$, Lemma 4.5 and Lemma 4.4 give $d_{\mathrm{blk}}(w, C_{n-j+1} \otimes \mathbb{K}) \leq \frac{M}{M-1}\delta$. Hence $1 + \eta \leq M/(M-1)$, i.e. $M \leq 1 + \eta^{-1}$. Lemma 3.6 transfers Δ to $\mathbb{K}$. □

5 The commitment scheme

Parameters. n (variables), k (redundancy), s (queries), $\mathbb{F} \subseteq \mathbb{K}$, a hash function H, and a public seed. The gate family $\mathcal{A}$ is derived from the seed with gates uniform on $\mathcal{G}^*(\mathbb{F})$. Merkle leaves are blocks: the leaf of a layer at ω' is the pair $(w(\omega', 0), w(\omega', 1))$.

5.1 Protocol

1. **Commit.** The prover computes $E_0 = \mathrm{Enc}_n(F) \in \mathbb{F}^{\Omega_n}$ and publishes its Merkle root R_0.

2. **Fold.** For $j = 1, \ldots, n$: the verifier sends $z_j \in \mathbb{K}$ uniform (in the non-interactive version $z_j = H(R_0, \ldots, R_{j-1})$). The prover computes $E_j = \mathrm{fold}_{j,z_j}(E_{j-1}) \in \mathbb{K}^{\Omega_{n-j}}$. For $j < n$ it publishes the root R_j. For $j = n$ it sends $E_n \in \mathbb{K}^{2^k}$ in full.

3. **Output.** The verifier checks that E_n is constant and outputs $y = E_n(\beta)$, the claimed value of $F(z_1, \dots, z_n)$.

4. **Query.** Repeat s times: sample $\omega = (\beta, \theta_n, \dots, \theta_1) \in \Omega_n$ uniformly. For $j = 1, \dots, n$, open the block of E_{j-1} at $\omega^{(j)}$ and the entry $E_j(\omega^{(j)})$; the latter lies in the block of E_j at $\omega^{(j+1)}$, opened in the next step, or in E_n. Check

$$E_j(\omega^{(j)}) = W_L(z_j)\,\tfrac{d}{\delta}\,E_{j-1}(\omega^{(j)}, 1) - W_R(z_j)\,\tfrac{c}{\delta}\,E_{j-1}(\omega^{(j)}, 0)$$

with $A_j(\omega^{(j)}) = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, together with all Merkle paths.

Per query: n block openings, Merkle paths of lengths $n + k - j$, and $O(n)$ field operations. Prover: $(n+1)2^{n+k}$ operations to encode and 2^{n+k} to fold, plus hashing.

6 Security

Theorem 6.1 (Completeness)**.** *An honest prover is always accepted, and $y = F(z_1, \dots, z_n)$.*

Proof. Lemma 4.2 and Lemma 2.4. □

For words x, x' on the same layer write $\mathrm{Agr}(x, x') = \{\omega : x(\omega) = x'(\omega)\}$, and for $w, c \in \mathbb{K}^{\Omega_i}$ $(i \geq 1)$ write $\mathrm{BAgr}(w, c) \subseteq \Omega_{i-1}$ for the set of blocks on which w and c agree in both entries.

Lemma 6.2 (Folding preserves agreement sets)**.** *Let $1 \leq j \leq n$ and $C = C_{n-j} \otimes \mathbb{K}$ with $\Delta(C) \geq \Delta$. Let $\varepsilon > 0$ and*

$$0 \leq \delta < \delta_{\max}(\Delta) := \min\Big(1 - (1-\Delta)^{1/3},\ \tfrac{\Delta}{2}\Big).$$

Put

$$\gamma = \frac{(1-\delta)^3 - (1-\Delta)}{\Delta}, \qquad M = \Big\lfloor \frac{\delta}{\Delta - 2\delta} \Big\rfloor + 2, \qquad N_1 = \Big\lceil \max\Big(\frac{12}{\Delta\gamma},\ \frac{2M}{\gamma} + 2\Big) \Big\rceil.$$

For every $w \in \mathbb{K}^{\Omega_{n-j+1}}$ there is a set $B(w) \subseteq \mathbb{K}$ with $|B(w)| < N_1 + 1/\varepsilon$ such that, for every $z \notin B(w)$ and every $c' \in C$ with $d_{\mathrm{pos}}(\mathrm{fold}_{j,z} w, c') \leq \delta$, there is $c \in C_{n-j+1} \otimes \mathbb{K}$ with

$$\mathrm{fold}_{j,z}(c) = c' \qquad \text{and} \qquad \big|\mathrm{Agr}(\mathrm{fold}_{j,z} w,\, c') \setminus \mathrm{BAgr}(w, c)\big| \leq \varepsilon\, L_{n-j}.$$

Proof. Write $L = L_{n-j}$ and $\mathrm{fold}_{j,z} w = A + zB$ (Lemma 4.4). Let $\mathcal{Z} = \{z : d_{\mathrm{pos}}(A + zB, C) \leq \delta\}$. Since $2\delta < \Delta$, each $z \in \mathcal{Z}$ has a unique codeword c_z within distance δ; let S_z be its agreement set, $|S_z| \geq (1-\delta)L$. Consider the points $P_z = (z, c_z)$ of the affine space $\mathbb{K} \times C$.

Step 1 (good triples are collinear). Let $z_1, z_2, z_3 \in \mathcal{Z}$ be distinct with $|S_{z_1} \cap S_{z_2} \cap S_{z_3}| > (1-\Delta)L$. Put $b = (c_{z_1} - c_{z_2})/(z_1 - z_2)$ and $a = c_{z_1} - z_1 b$. On $S_{z_1} \cap S_{z_2}$ we have $(A, B) = (a, b)$. On the triple intersection, $c_{z_3} = A + z_3 B = a + z_3 b$. Two codewords agreeing on more than $(1-\Delta)L$ positions are equal, so $c_{z_3} = a + z_3 b$: the three points lie on one line.

Step 2 (many good triples). Let $N = |\mathcal{Z}| \geq 3$ and $Z_p = \{z \in \mathcal{Z} : p \in S_z\}$. Then $\sum_p |Z_p| \geq N(1-\delta)L$, and the sum of $|S_{z_1} \cap S_{z_2} \cap S_{z_3}|$ over ordered triples of distinct elements equals $\sum_p |Z_p|(|Z_p|-1)(|Z_p|-2)$. Since $x(x-1)(x-2) \geq \max(0, x-2)^3$ on integers and the right side is convex, this sum is at least $L\big(N(1-\delta)-2\big)^3$. Dividing by N^3, the average triple intersection

is at least $L(1-\delta-2/N)^3$. Each triple intersection is at most L, and at most $(1-\Delta)L$ for a bad triple. So the fraction of good triples is at least

$$\gamma_N = \frac{(1-\delta-2/N)^3-(1-\Delta)}{\Delta} \;\geq\; \gamma - \frac{6}{N\Delta}.$$

Step 3 (a heavy line). Distinct points of $\mathbb{K}\times C$ span a unique line. If line ℓ carries m_ℓ of the points, ordered collinear triples number $\sum_\ell m_\ell(m_\ell-1)(m_\ell-2) \leq (\max_\ell m_\ell)\,N(N-1)$. Hence some line carries at least $\gamma_N(N-2)$ points. For $N \geq N_1$ this is at least M.

Step 4 (agreement along a line). Let a line carry $m \geq 2$ points, i.e. $c_z = a + zb$ for m values z. Put $E = \{p : (A,B)(p) \neq (a,b)(p)\}$. As in Lemma 4.5, each $p \in E$ lies in at most one S_z, while $|E \setminus S_z| \leq \delta L$. Hence $|E| \leq \frac{m}{m-1}\delta L$.

Step 5 (absorption). Let a line ℓ with pair (a,b) carry $m \geq M$ points, and let E be as in Step 4, so $|E| \leq \frac{M}{M-1}\delta L$. For any $z \in \mathcal{Z}$, the codewords c_z and $a+zb$ both equal $A+zB$ on $S_z \setminus E$, a set of size at least $\big(1-\delta-\frac{M}{M-1}\delta\big)L$. Since $M-1 > \frac{\delta}{\Delta-2\delta}$, we have $\delta\big(2+\frac{1}{M-1}\big) < \Delta$, so this size exceeds $(1-\Delta)L$ and $c_z = a+zb$. Hence *every* point P_z, $z \in \mathcal{Z}$, lies on ℓ.

Step 6 (the exceptional set).

- If $N < N_1$, put $B(w) = \mathcal{Z}$.
- Otherwise Step 3 gives a line ℓ carrying at least M points, with pair (a,b) and set E, and Step 5 puts all of $\mathcal{Z}$ on ℓ.
- On ℓ we have $S_z = E^c \cup (E \cap S_z)$ and $\sum_{z\in\ell} |E \cap S_z| \leq |E| \leq L$. So fewer than $1/\varepsilon$ values $z \in \ell$ have $|E \cap S_z| > \varepsilon L$.
- Put $B(w) = \{z \in \mathcal{Z} : |E \cap S_z| > \varepsilon L\}$.

For $z \notin B(w)$ with a codeword c' within δ, we have $z \in \ell$ and $c' = c_z = a + zb$. Let c be the word of $C_{n-j+1} \otimes \mathbb{K}$ built from (a,b). Then $\mathrm{fold}_{j,z}c = c'$, and $\mathrm{BAgr}(w,c) = E^c$ by Lemma 4.4. □

Remark 6.3. For $\Delta < 3-\sqrt{5} \approx 0.76$ the binding term is the triple radius $1-(1-\Delta)^{1/3}$, which exceeds $\Delta/3$. Example: for $\Delta = 0.508$ the radius grows from 0.169 to 0.21.

Fix a family with $\Delta(C_i) \geq \Delta^*$ for all i, a threshold $0 < \delta < \delta_{\max}(\Delta^*)$ (Lemma 6.2) and $\varepsilon > 0$. Say the commitment R_0 *binds* $F^* \in \mathrm{ML}(x_1,\ldots,x_n)$ (coefficients in $\mathbb{K}$) if $d_{\mathrm{blk}}(E_0, \mathrm{Enc}_n(F^*)) < \delta$. Such an F^* is unique, because $2\delta < \Delta^*$.

Theorem 6.4 (Soundness)**.** *For any prover, the probability that the verifier accepts while either R_0 binds no polynomial, or R_0 binds F^* and $y \neq F^*(z_1,\ldots,z_n)$, is at most*

$$\frac{n\,(N_1+1/\varepsilon)}{|\mathbb{K}|} \;+\; \big(1-\delta+n\varepsilon\big)^s$$

in the IOP model (Merkle binding assumed).

Proof. Write $f_j = \mathrm{fold}_{j,z_j}(E_{j-1})$; the check of step j on a query passes iff $E_j(\omega^{(j)}) = f_j(\omega^{(j)})$.

Good challenges. E_{j-1} is fixed before z_j is drawn. Let G be the event $z_j \notin B(E_{j-1})$ for all j (Lemma 6.2). Then $\Pr[\neg G] \leq n(N_1+1/\varepsilon)/|\mathbb{K}|$. Assume G, and that the verifier checked that E_n is the constant y.

Descent. Put $c'_n = E_n \in C_0 \otimes \mathbb{K}$. For $j = n, n-1, \ldots, 1$:

- if $d_{\mathrm{pos}}(f_j, c'_j) > \delta$, stop and set $J = j$;

- otherwise, Lemma 6.2 gives c'_{j-1} with $\mathrm{fold}_{j,z_j} c'_{j-1} = c'_j$ and a set $X_j = \mathrm{Agr}(f_j, c'_j) \backslash \mathrm{BAgr}(E_{j-1}, c'_{j-1})$ with $|X_j| \le \varepsilon L_{n-j}$.

If the descent never stops, set $J = 0$. The codewords c'_j and sets X_j depend only on the transcript before the query phase.

One query. Let ω be a query and $p_j = \omega^{(j)}$. Suppose it passes every check and $p_j \notin X_j$ for all $j > J$. We show by descending induction that $E_j(p_j) = c'_j(p_j)$ for $j \ge J$.

- For $j = n$ this is the definition of c'_n.
- Assume it for $j > J$. The check at step j gives $f_j(p_j) = E_j(p_j) = c'_j(p_j)$, so $p_j \in \mathrm{Agr}(f_j, c'_j)$. Since $p_j \notin X_j$, the block of E_{j-1} at p_j agrees with c'_{j-1} in both entries. In particular $E_{j-1}(p_{j-1}) = c'_{j-1}(p_{j-1})$.

Consequently:

- if $J \ge 1$, then $p_J \in \mathrm{Agr}(f_J, c'_J)$, a set of relative size $< 1 - \delta$;
- if $J = 0$, the block of E_0 at p_1 lies in $\mathrm{BAgr}(E_0, c'_0)$, a set of relative size $1 - d_{\mathrm{blk}}(E_0, c'_0)$.

Since each p_j is uniform on its layer,

$$\Pr[\text{query passes}] \le n\varepsilon + \max\big(1 - \delta,\ 1 - d_{\mathrm{blk}}(E_0, c'_0)\big).$$

The bad cases. Folding c'_0 through all rounds gives c'_n, the constant y. Write $c'_0 = \mathrm{Enc}_n(\widetilde{F})$; Lemma 4.2 gives $y = \widetilde{F}(z)$.

- If R_0 binds no polynomial, then $d_{\mathrm{blk}}(E_0, c'_0) \ge \delta$.
- If R_0 binds F^* and $y \ne F^*(z)$, then $\widetilde{F} \ne F^*$. Distinct codewords are at block distance $\ge \Delta^*$, so $d_{\mathrm{blk}}(E_0, c'_0) \ge \Delta^* - \delta > \delta$.

In both cases a query passes with probability $\le 1 - \delta + n\varepsilon$. The s query positions are independent of everything fixed above. □

Remark 6.5 (Why this bound has no factor n)**.** The argument follows the path of a query from the constant y down to T_0. Lemma 6.2 guarantees that whenever a check passes on a position, the whole pair below it, including the sibling not on the path, agrees with one fixed valid table. A layer-by-layer distance count cannot see this, and loses a factor n.

6.1 Committing only some layers

Folding is a public deterministic map, so a layer that is not committed can be recomputed by the verifier from the layer above it. This gives folding by 2^m per round with no new soundness loss.

Definition 6.6 (Commitment schedule)**.** A *schedule* is a set $J \subseteq \{1, \dots, n-1\}$ of layers that are *not* committed. The protocol Π_J is the protocol of §5 with these changes.

- For $j \in J$ the prover sends no root R_j. The challenge z_{j+1} is sampled right after z_j, with no prover message in between (non-interactively: $z_{j+1} = H(\text{transcript}, z_j)$).

- In the query phase, for a committed layer $j_0 \notin J$ followed by the run $j_0+1, \ldots, j_0+m-1 \in J$ and the next committed layer $j_0+m \notin J$ (or $j_0+m=n$), the verifier opens the whole group of 2^m entries of E_{j_0} below the prefix $\omega^{(j_0+m)}$, with one Merkle path. It recomputes the values of the skipped layers on this group with the gate rule, and checks the resulting value against the opened entry of E_{j_0+m} (or against E_n).

Merkle leaves of layer j_0 are groups of 2^m entries. $J=\emptyset$ is the protocol of §5; $J=\{1,3,5,\ldots\}$ is folding by 4 per round.

Theorem 6.7 (Soundness with any schedule)**.** *For every J, the soundness error of Π_J is at most the bound of Theorem 6.4:*

$$\frac{n\,(N_1+1/\varepsilon)}{|\mathbb{K}|} \;+\; \big(1-\delta+n\varepsilon\big)^s.$$

Proof. Work in the IOP model of Theorem 6.4: a root is an oracle to the full table, so the grouping of Merkle leaves plays no role. Let P^* be any prover for Π_J. Build a prover P' for $\Pi_\emptyset$:

- P' runs P^* and forwards its messages and roots.
- For $j \in J$, after receiving z_j, P' defines $E_j := \mathrm{fold}_{j,z_j}(E_{j-1})$, where E_{j-1} is the committed table, or the table defined in the same way if $j-1 \in J$. It commits to E_j honestly and sends the root R_j.
- In the query phase P' answers the openings of committed layers with the answers of P^*, and the openings of the layers $j \in J$ honestly from E_j.

Same challenges. In both protocols each z_j is uniform on $\mathbb{K}$ and independent of everything before it. Inserting the root R_j ($j \in J$) does not change this, because E_j and hence R_j are functions of messages already sent.

Same acceptance. Fix a query leaf ω and a run $j_0, \ldots, j_0+m$ as in the definition.

- The blocks of E_{j_0} that $\Pi_\emptyset$ opens along ω in this run are contained in the group that Π_J opens, and the Π_J verifier sees them under one Merkle path to R_{j_0}.
- For $j \in J$, the $\Pi_\emptyset$ check at layer j compares $E_j(\omega^{(j)})$ with the gate rule applied to a block of E_{j-1}. Since $E_j = \mathrm{fold}_{j,z_j}(E_{j-1})$ entrywise, it passes.
- The $\Pi_\emptyset$ check at the committed layer j_0+m compares the opened entry of E_{j_0+m} with the gate rule applied to a block of E_{j_0+m-1}. That block is the recomputed value of the group, so this is exactly the Π_J check.

Hence the $\Pi_\emptyset$ verifier accepts against P' whenever the Π_J verifier accepts against P^*, on the same randomness, with the same output y and the same root R_0. Theorem 6.4 bounds the probability of the bad event for P'. □

Remark 6.8 (Costs)**.** Committing every m-th layer gives $\lceil n/m \rceil$ Merkle trees, and per query $\lceil n/m \rceil$ paths, each with a leaf of 2^m entries. The verifier work for the gate rule is unchanged, about 2^m-1 gate evaluations per committed layer. The same argument covers sending the last r layers in full: the verifier folds E_{n-r} itself and checks that the result is constant. The query count s and the challenge field are unchanged.

6.2 Batch opening

To open t polynomials $f_1, \dots, f_t \in \mathrm{ML}(x_1, \dots, x_n)$ at the same random point, the prover runs one protocol on the t-tuple of tables.

Protocol. Each position carries the tuple $\big(E_j^{(1)}(\omega), \dots, E_j^{(t)}(\omega)\big)$, where $E_0^{(i)} = \mathrm{Enc}_n(f_i)$. One Merkle tree per committed layer has these tuples (grouped in blocks) as leaves. All t tables are folded with the same challenges z_j. The last layer is sent for every i; the verifier checks that each component is constant and outputs $y_i = E_n^{(i)}$. A query opens one path per committed layer and checks the gate rule in every component. The transcript absorbs the whole last layer before the query positions are derived. No random linear combination of the f_i is used.

For a linear code $C \subseteq \mathbb{K}^L$ let $C^{(t)} = \{(c_1, \dots, c_t) : c_i \in C\}$, viewed as a code over the alphabet $\mathbb{K}^t$: a position is nonzero, or two words agree there, according to the whole t-tuple. Distances, Agr and BAgr are taken in this sense.

Proposition 6.9. $\Delta(C^{(t)}) = \Delta(C)$.

Proof. A nonzero $(c_1, \dots, c_t)$ has a nonzero component c_i, and its support contains that of c_i, so its weight is at least $\Delta(C)L$. The word $(c^\star, 0, \dots, 0)$ with $c^\star$ of minimum weight shows equality. □

Lemma 6.10. *Lemma 6.2 holds verbatim for $C^{(t)}$ in place of C, with the same γ, M, N_1 and the same bound $|B(w)| < N_1 + 1/\varepsilon$, independent of t.*

Proof. Folding acts componentwise with the same challenge and gates, so $\mathrm{fold}_{j,z} w = A + zB$ with A, B taking values in $\mathbb{K}^t$, and the line form (Lemma 4.4) holds componentwise. We check each step of the proof of Lemma 6.2.

- Unique decoding and Steps 1, 5: they use only that $C^{(t)}$ is a $\mathbb{K}$-linear space and that two codewords agreeing on more than $(1-\Delta)L$ positions are equal, which holds by Proposition 6.9.
- Steps 2–3: pure counting over positions and over points of the affine space $\mathbb{K} \times C^{(t)}$.
- Step 4: if $p \in E$ lay in $S_z \cap S_{z'}$, subtracting the two tuple equalities gives $B(p) = b(p)$ and $A(p) = a(p)$ in $\mathbb{K}^t$, a contradiction; the count is unchanged.
- Step 6: write $a = (a_i)$, $b = (b_i)$ with $a_i, b_i \in C$ and lift each $a_i + zb_i$ with the line form; the tuple c of lifts satisfies $\mathrm{fold}_{j,z} c = c_z$ and $\mathrm{BAgr}(w, c) = E^c$. □

Theorem 6.11 (Soundness of batch opening)**.** *Say that the commitment binds $(f_1^\star, \dots, f_t^\star)$ if $d_{\mathrm{blk}}\big(E_0, (\mathrm{Enc}_n(f_i^\star))_i\big) < \delta$. For any prover, the probability that the verifier accepts while the commitment binds no tuple, or binds $(f_i^\star)$ but $y_i \neq f_i^\star(z_1, \dots, z_n)$ for some i, is at most*

$$\frac{n\,(N_1 + 1/\varepsilon)}{|\mathbb{K}|} + \big(1 - \delta + n\varepsilon\big)^s.$$

In particular the number of queries does not depend on t. The same holds for every commitment schedule.

Proof. Repeat the proof of Theorem 6.4 with tuple-valued tables, using Lemma 6.10 once per round (not once per component), so the good-challenge event costs $n(N_1 + 1/\varepsilon)/|\mathbb{K}|$. A check passes iff the tuples are equal, so the descent and the one-query induction are unchanged. In the bad cases, if the descent reaches $J = 0$ then $c_0' = (\mathrm{Enc}_n(\tilde{f}_i))_i$ with $y_i = \tilde{f}_i(z)$, so $(\tilde{f}_i) \neq (f_i^\star)$ and, by uniqueness of the bound tuple (Proposition 6.9 and $2\delta < \Delta^*$), $d_{\mathrm{blk}}(E_0, c_0') \geq \delta$. For schedules, the reduction of Theorem 6.7 recomputes skipped layers componentwise. □

Remark 6.12 (Cost)**.** The number of queries and Merkle paths is that of a single opening; only the leaves grow, by a factor t in field elements. The prover encodes and folds t tables.

Remark 6.13 (Non-interactive version)**.** With $z_j = H(\text{transcript up to } R_{j-1})$ and query positions hashed from the full transcript, a prover making Q hash queries gains at most a factor Q on the first term, plus the Merkle collision term.

Remark 6.14 (Concrete parameters, proven)**.** Take $\mathbb{F} = \mathbb{F}_{2^{31}-1}$, $n = 20$, $k = 6$ (64 copies, $\Delta^* \geq 0.508$), $\mathbb{K}$ of degree 5 over $\mathbb{F}$, $\delta = 0.20$ (so $\gamma \approx 0.0394$, $M = 3$, $N_1 = 600$) and $\varepsilon = 2^{-20}$. Then $n(N_1 + 1/\varepsilon)/|\mathbb{K}| \leq 2^{-130}$, and $s = 398$ queries give $(1 - \delta + n\varepsilon)^s < 2^{-128}$.

Remark 6.15 (Any field)**.** Nothing above uses structure of $\mathbb{F}$ beyond its size: no subgroup and no special element. In particular every result holds in characteristic 2: the proofs use only $t^0 \neq t^1$, division by $t^1 - t^0$, and Schwartz–Zippel. In a binary field, additions are XOR and Lemma 2.5 reads $\sum_{x\in\{0,1\}} F = c\,F_L + d\,F_R$. If $\mathbb{F}$ is too small for Theorem 3.10 (roughly $q < 2^{30}$ at $n = 20$), sample the gates in an extension $\mathbb{F}' \supseteq \mathbb{F}$. The table E_0 then lives in $\mathbb{F}'$, and every statement holds with $q = |\mathbb{F}'|$. The challenge field $\mathbb{K}$ only needs $n(N_1 + 1/\varepsilon)/|\mathbb{K}|$ to be negligible.

Open problem 6.16 (Up to $\Delta^*/2$)**.** Remove the triple-radius condition in Lemma 6.2, so that $\delta < \Delta^*/2$ suffices, with a bad set of size $O(L) + O(1/\varepsilon)$. At $n = 20$, four copies, $i_0 = 3$ and $g = 16$ this lowers the query count from 407 to 353. Pairs of close challenges already define lines up to radius $\Delta/2$, and the absorption step (Step 5 of Lemma 6.2) holds there; what is missing is the existence of a heavy line, which for general linear codes in this range is an open question.

Open problem 6.17 (Correlated agreement up to the Johnson radius)**.** Let $\delta < 1 - \sqrt{1 - \Delta}$ and $\text{fold}_r w = A + rB$. Call a pair (r, c') with c' a codeword at distance $\leq \delta$ from $A + rB$ *good* if some c in the upper code satisfies $\text{fold}_r c = c'$ and $|\text{Agr}(A + rB, c') \setminus \text{BAgr}(w, c)| \leq \varepsilon L$. Show that the challenges having a pair that is not good number at most $c_1 L + c_2/\varepsilon$.

Evidence. Exhaustive list decoding of SPARK codes over GF(32), GF(64) and GF(256) (length 16, lists of size up to 3, 1000 adversarial and planted trials per code and radius) found at most 2 such challenges for $\varepsilon L = 1$ and at most 1 for $\varepsilon L = 2$, far below the allowance $1/\varepsilon$. Combined with Theorem 6.4, a proof would lower the query count at the parameters above to about 300.

Remark 6.18 (Structure of the minors of C_2)**.** Write a coordinate row of C_2 as $(1, y_1, y_2, y_1 y_2)$, with y_1 the point of the first gate and y_2 that of the child gate. For one copy the single maximal minor is $\pm(t^1_\varnothing - t^0_\varnothing)^2(t^1_0 - t^0_0)(t^1_1 - t^0_1)$. For two copies the 70 maximal minors fall into three classes by how the four lower blocks are used: 6 of type $(2, 2)$ and 48 of type $(2, 1, 1)$ factor into differences of structural points, while the 16 of type $(1, 1, 1, 1)$, one position from each lower block, are a single degree-4 relation among 8 points that does not factor into point differences (checked over $\mathbb{Q}$, and over GF(2^8) against all 28 candidate differences). Hence pairwise distinctness of gate points certifies every minor containing a double block, but not the all-single class, whose size grows like 2^{D_i}. Certifying C_4 or C_5 cheaply therefore needs a new idea; it would raise Δ^* to 0.634 or 0.643 at the parameters above.

7 Distance in practice

Theorem 3.14 loses about $n\,H_2(2^{-k})/\log_2 q$ in total. At realistic field sizes the experiments below found no codeword below the Singleton limit; whether a loss growing with n is intrinsic is open (Problem 3.24).

Experiment. Over $\mathbb{F}_{2^{31}-1}$, with per-node random gates:

- **Rank tests.** Test sets of 2^n positions for linear dependence of their columns, both uniformly random sets and unions of whole subtrees, where the tree structure would show a weakness.
- **Zero search.** Choose $2^n - 1$ positions, take the nonzero polynomial vanishing there, and count all its zeros. More than $2^n - 1$ zeros would beat the Singleton limit.

n	copies	length	2^n	singular (random)	singular (subtrees)	max zeros	Δ found
6	2	128	64	0/300	0/300	63	0.508
6	4	256	64	0/300	0/300	63	0.754
6	8	512	64	0/300	0/300	63	0.877
8	2	512	256	0/120	0/120	255	0.502
8	4	1024	256	0/120	0/120	255	0.751
8	8	2048	256	0/120	0/120	255	0.876

Table 6: Every test meets the Singleton limit exactly. Control with per-layer gates ($n = 6$): 225 to 300 singular sets out of 300, and codewords of relative weight 2^{-6}, as Proposition 3.7 predicts.

setting	copies	Δ	queries	size of E_0	status
gates in Goldilocks	64	0.811	183	$6.7 \cdot 10^7$	proven (Thm. 3.14)
gates in Goldilocks	16	0.579	327	$1.7 \cdot 10^7$	proven (Thm. 3.14)
gates in Goldilocks^3	8	0.812	183	$8.4 \cdot 10^6$ in $\mathbb{F}_{p^3}$	proven (Thm. 3.14)
gates in M31	64	0.508	398	$6.7 \cdot 10^7$	proven (Thm. 3.10)
any field, $q \geq 2^{30}$	8	≈ 0.86	≈ 163	$8.4 \cdot 10^6$	if the level count is removed

Table 7: Parameters for $n = 20$ at 2^{-128} soundness (Theorem 6.4).

7.1 Specialized gates for the binary-field implementation

For the production binary-field implementation we specialize each parent gate to $t^0 = t$, $t^1 = t+1$. In characteristic two these points are distinct deterministically. For inputs u, v,

$$a = u + t\,v, \qquad b = a + v = u + (t+1)v.$$

The fold becomes

$$w^0 + (z + t)(w^1 + w^0),$$

because $t^1 - t^0 = 1$.

The Theorem 3.19 distance argument is unchanged. Double-parent constraints force $v = 0$ and then $u = 0$. A singleton equation $u + (t + b)v = 0$ has exactly one solution t when $v \neq 0$, hence probability $1/q$, and parents remain independent.

At $n = 20, k = 2, i_0 = 3, s = 407$, median prover time falls by 17.4% with one Rayon thread and 14.5% with eight. Encoding multiplications fall exactly from 83,886,080 to 41,943,040; folding removes 8,388,600 base-field multiplications; commitment remains exactly 4,793,481 Merkle hashes.

The pre-deduplication opening structure is unchanged. The deterministic v0.10 proof is 592,577 bytes, 0.27% larger than v0.9 because domain separation changes query positions and therefore Merkle-path deduplication.

Linear proximity gap. For the optional GKL branch, Gao–Kan–Li applies for

$$\delta \le 1 - (1 - \Delta_C + \eta)^{1/3}$$

and bounds the exceptional combining points by

$$L + \frac{6}{\eta} + \frac{2}{\eta((1 - \Delta_C + \eta)^{1/3} - \sqrt{1 - \Delta_C + \eta})}.$$

This is the 1.5-Johnson result, not a full Johnson-radius claim.

The $k = 3$ record. The currently landable $k = 3$ point is $i_0 = 2, s = 277$. The $s = 261$ value is conditional on a checked $C_3 = [64, 8]$.

Transparent checked setup. The fixed domain-separated seed and first accepted public counter are immutable protocol constants after the finite checked-prefix test. The checked-setup failure budget is a one-time transparent-setup event, not a per-proof charge.

7.2 Leaf layout and hashing cost

With a commitment schedule, the query phase never reads a single block of a committed layer in isolation. For a committed layer j_0 followed by a run of length m, every query opens the whole group of 2^m entries of E_{j_0} below the prefix $\omega^{(j_0+m)}$ (Definition of a commitment schedule). The natural Merkle leaf is therefore this group. A tree with one leaf per pair hashes the same data, but it also builds the $2^{m-1} - 1$ internal nodes inside every group. Their digests are never sent, since the verifier recomputes them from the opened entries.

Hash count. Let a committed layer have L entries of σ bytes, and let $\kappa(\ell) = \lceil \ell/64 \rceil$ be the number of BLAKE3 compressions for an ℓ-byte input ($\ell \le 1024$). With domain tags of τ_{leaf}, τ_{grp} and τ_{node} bytes, the two layouts cost

$$C_{\text{pair}} = \tfrac{L}{2}\,\kappa(\tau_{\text{leaf}} + 2\sigma) + \big(\tfrac{L}{2} - 1\big)\kappa(\tau_{\text{node}} + 64), \qquad C_{\text{grp}} = \tfrac{L}{2^m}\,\kappa(\tau_{\text{grp}} + 2^m\sigma) + \big(\tfrac{L}{2^m} - 1\big)\kappa(\tau_{\text{node}} + 64)$$

compressions. With tags written into the input ($\tau_{\text{leaf}} = \tau_{\text{node}} = 22$, $\tau_{\text{grp}} = 21$), an internal node costs $\kappa(86) = 2$ compressions.

Keyed hashing. BLAKE3 has a keyed mode whose key enters the chaining value and costs no input bytes. Taking the domain tag as the key, $K_{\text{leaf}} = \text{derive_key}(\text{“SPARK-BINARY v0.10 Merkle leaf”})$ and $K_{\text{node}} = \text{derive_key}(\text{“SPARK-BINARY v0.10 Merkle node”})$, gives $\tau_{\text{grp}} = \tau_{\text{node}} = 0$. An internal node is then one compression of its two 32-byte children, and a group leaf costs $\kappa(2^m\sigma)$.

For the first layer ($\sigma = 16$) and $m = 3$, one group costs $4 \cdot 1 + 3 \cdot 2 = 10$ compressions with pair leaves, $\kappa(149) = 3$ with a tagged group leaf and $\kappa(128) = 2$ with a keyed group leaf. Over all committed layers at $n = 20$, $k = 2$:

m	pair leaves	group leaves, tagged	group leaves, keyed	change
2	9,087,628	5,941,904	3,145,714	−65%
3	7,489,810	3,145,718	1,947,349	−74%
4	6,850,678	2,027,234	1,467,999	−79%

The keyed group layout is the one used from v0.10 on.

Proof and verifier. The opened field elements are the same in both layouts. Each authentication path is $m-1$ levels shorter, but those levels lie inside the group and were already recomputed by the verifier. So the proof changes only through the deduplication of shared paths, which depends on the query positions; in our measurements the change stays within $\pm 2\%$. The verifier hashes one leaf per group instead of 2^{m-1} leaves and $2^{m-1}-1$ nodes.

Security. The proof of Theorem 6.7 works in the IOP model, where a root is an oracle to the whole table and the grouping of leaves plays no role. Merkle binding holds for any fixed public grouping. Leaves and internal nodes are hashed under distinct keys derived from distinct context strings, so a leaf cannot be confused with a node. Keyed BLAKE3 is a PRF under its key and is collision resistant for a fixed public key, which is all Merkle binding needs. The format byte separates the layouts in the transcript. The soundness bound, the query count s and the challenge field are unchanged.

Measurements. Table 8 reports the commitment phase alone, with tagged leaves. It builds exactly the trees of the prover on the same tables, with the layouts interleaved in one process. Table 9 reports full proofs for all four combinations of gate family and leaf layout. The time

n	m	pair leaves (ms)	group leaves (ms)	change	compressions
16	3	59.2	27.9	−53%	468,100 → 196,604
18	3	236.9	111.2	−53%	1,872,436 → 786,416
20	2	1139.9	799.5	−30%	9,087,628 → 5,941,904
20	3	944.8	442.6	−53%	7,489,810 → 3,145,718
20	4	864.9	302.4	−65%	6,850,678 → 2,027,234

Table 8: Commitment phase, $k = 2$, BLAKE3, one thread, Apple M1, median of 20 interleaved repetitions; the interquartile ranges do not overlap.

follows the compression count: about 126 ns per compression with pair leaves and 135–150 ns with group leaves on one thread. The relative changes are the same on an Intel Xeon at 2.8 GHz (−30%, −56%, −65% at $n = 20$). Keyed hashing removes a further 30% of the commitment time at $m = 3$ (295 → 205 ms) and 22% at $m = 4$ (214 → 167 ms) on the Xeon, one thread, median of 6 interleaved repetitions.

	$m = 3$			$m = 4$		
gates, leaves	prover	verifier	proof	prover	verifier	proof
random, pair	1669 ms	10.6 ms	588,170 B	1567 ms	13.4 ms	655,682 B
$(t, t+1)$, pair	1367 ms	10.2 ms	594,218 B	1305 ms	12.8 ms	659,618 B
random, group	1137 ms	9.0 ms	593,418 B	1001 ms	10.6 ms	661,314 B
$(t, t+1)$, group	**882 ms**	8.4 ms	585,706 B	**738 ms**	9.8 ms	658,658 B

Table 9: Full proofs at $n = 20$, $k = 2$, $s = 407$, $g = 16$, one thread, Apple M1. All four configurations are built from the same commit and measured in one session, cycling through them in every round (median of 20 runs each). The first row has the cost of v0.9.1 (random gates, pair leaves). Group leaves are tagged here; the keyed layout of v0.10 is faster still.

The two optimizations are independent and their gains multiply. Specialized gates (§7.1) shorten encoding and folding. Group leaves shorten the commitment. Together they make the single-thread prover 1.89× faster at $m = 3$ (1669 → 882 ms) and 2.12× faster at $m = 4$ (1567 → 738 ms). The verifier gets faster too, and the proof size stays within ±1% of the baseline for each schedule. With eight threads the medians are 510 → 323 ms at $m = 3$ and 462 → 176 ms at $m = 4$; the run-to-run spread is larger there.

The two schedules serve different needs. The *compact* profile $m = 3$ gives the smallest proof (586 KB). The *fast* profile $m = 4$ gives the fastest prover (738 ms on one thread, 176 ms on eight) for a proof 12% larger. Both have the same soundness bound and query count.

8 Implementation and parameters

Instantiation. Gates in $\mathbb{F} = \mathrm{GF}(2^{128})$ with modulus $x^{128} + x^7 + x^2 + x + 1$ (carry-less multiplication); challenges and folded layers in $\mathbb{K} = \mathbb{F}[y]/(y^2 + y + \alpha)$ with $\mathrm{Tr}(\alpha) = 1$, so $|\mathbb{K}| = 2^{256}$; hash BLAKE3; gates derived from a public seed by a keyed hash, one compression per gate, in the random-oracle model; checked setup at $i_0 = 3$ (all $\binom{32}{8}$ minors of C_3, done once).

Security. Target 128 bits against $Q = 2^{64}$ hash queries, i.e. each term of Theorem 6.4 at most 2^{-192}. At $n = 20$ and four copies: $\Delta^* = 0.6157$, $\delta = 0.2593$, $M = 4$, $N_1 = 545$, $\varepsilon = 2^{-59.67}$; fold term 192.0 bits; with $g = 16$ bits of grinding, $s = 407$ queries give a query term of 192.3 bits. These values do not depend on the commitment schedule (Theorem 6.7) nor on the batch size (Theorem 6.11).

n	s	configuration	prover	prover (8 thr.)	verifier	proof	per polynomial
16	376	single, $m = 3$	162 ms	31.7 ms	6.6 ms	288 KiB	—
		single, $m = 4$	124 ms	24.7 ms	8.5 ms	329 KiB	—
		batch $t = 16$, $m = 3$	875 ms	255 ms	34.3 ms	2.98 MiB	54.7 ms, 2.1 ms, 191 KiB
18	391	single, $m = 3$	449 ms	112 ms	9.0 ms	429 KiB	—
		single, $m = 4$	437 ms	115 ms	11.2 ms	471 KiB	—
		batch $t = 16$, $m = 3$	3.73 s	1.26 s	46.0 ms	3.99 MiB	233 ms, 2.9 ms, 255 KiB
20	407	single, $m = 3$	1.82 s	0.49 s	11.4 ms	580 KiB	—
		single, $m = 4$	1.72 s	0.46 s	14.5 ms	644 KiB	—
		batch $t = 16$, $m = 3$	17.1 s	5.87 s	59.9 ms	5.26 MiB	1.07 s, 3.7 ms, 336 KiB

Table 10: Frozen v0.8 benchmark (tag `v0.8`, commit `b8e8081`): four copies, $i_0 = 3$, $g = 16$, Apple M1 (16 GiB), median of 10 runs, every run verified. Prover and verifier single-threaded unless noted; the per-polynomial column gives prover, verifier and proof size divided by t. s is derived from Theorem 6.4 at each n for the 192-bit target. Not included: the one-time setup, 0.68 s of gate derivation plus 4.4 s for the C_3 check at $n = 20$. Peak memory at $n = 20$: 0.75 GB for a single opening, 3.2 GB for the batch.

9 Further algebraic facts

All statements below were checked numerically over $\mathbb{F}_{2^{31}-1}$ (Rust test suite).

1. **Kronecker form.** Let the bottom table T_0 come from per-layer gates $A_1, \dots, A_n$, where layer i splits x_{n-i+1} and children are ordered (F_L, F_R). With the coefficient vector α indexed so that bit $n-1$ corresponds to x_n,
$$\alpha = (A_1 \otimes A_2 \otimes \cdots \otimes A_n)\, T_0$$
(A_1 acting on the most significant index bit), and $\det(\bigotimes_i A_i) = \prod_i \det(A_i)^{2^{n-1}}$.

2. **Sum row.** With $r(A) = [\,2a + c,\ 2b + d\,]$, $\sum_{x \in \{0,1\}^n} F(x) = \big(r(A_1) \otimes \cdots \otimes r(A_n)\big) T_0$.

3. **Representation.** $\rho : \mathrm{GL}_2(\mathbb{F})^n \to \mathrm{GL}(\mathbb{F}^{2^n})$, $\rho(A_1, \dots, A_n) = \bigotimes_i A_i$, is a homomorphism with
$$\ker \rho = \Big\{ (\lambda_1 I, \dots, \lambda_n I) :\ \textstyle\prod_i \lambda_i = 1 \Big\}.$$
The folding map $\alpha \mapsto T_0$ is $\rho(A)^{-1} = \rho(A^{-1})$. The assignment $A \mapsto \rho(A)^{-1}$ is an anti-homomorphism. The homomorphism to use is $A \mapsto \rho(A^{-\top})$.

4. **Identity gate.** For $A_i = I$, T_0 is the coefficient vector with index bits reversed relative to the splitting order.